ACID Properties

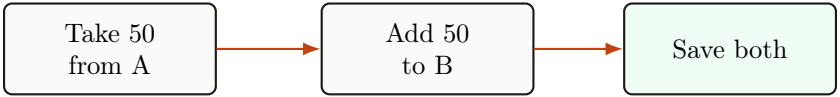
Analysis and Assessment of Design Quality

Atomicity = all or nothing



A transaction is one unit. No half-finished updates.

Bank transfer — the idea



Two writes. One business action.

Bank transfer — if it crashes

Bad	Atomic
A lost 50	crash → undo
B got nothing	A and B unchanged

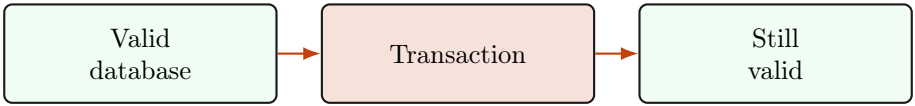
Money must not disappear halfway.

Same idea — place an order



Any step fails → undo all three

Consistency = rules stay true



After commit, your rules still hold.

Examples of rules

Balance cannot go below 0

Order must point to a real user

Stock cannot go negative

One seat → one booking

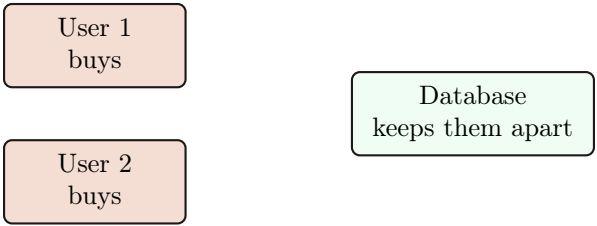
The database (and your app) enforce these.

Note on the word “consistency”

In ACID: rules / constraints hold.

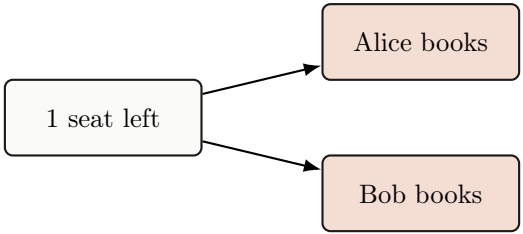
(CAP uses the same word for a different idea — ignore it for now.)

Isolation = concurrent work does not clash



Two people update the same data at once — isolation decides what is safe.

Last seat problem



Both must not get the seat.

Weak vs strong isolation

Too weak
Both get the seat

Strong enough
One wins, one fails

Stronger isolation = safer, but slower under load.

Isolation levels — plain view

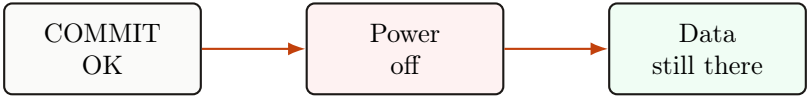
Weaker — faster, more surprises possible

Common default: only see committed data

Strongest — as if one tx ran at a time

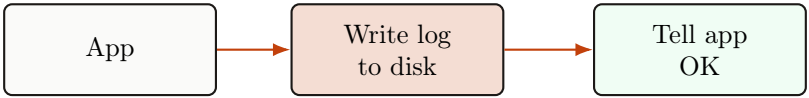
Pick stronger levels for money and scarce items.

Durability = committed data survives a crash



If the DB said “saved”, a reboot must not erase it.

How? Write the log first



WAL = write-ahead log (diary of changes before the main files).
Flush to disk before saying “committed”.

Safer vs faster

Strong ACID

Safer money paths
Slower under load

Relaxed

Higher throughput
More edge cases

Where to relax

OK to relax

Feeds, search, approximate counters

Do not relax

Money, stock, seats, permissions

Quiz — last seat

Quiz

Alice and Bob book the last seat at the same time. What do you need?

1. Hope they click at different times
2. A rule (or lock) so only one booking can win
3. A faster server
4. Turn off saving to disk

Pause — discuss with a neighbour or answer privately before the next slide.

Answer — last seat

Answer

2. A unique seat rule and/or strong enough isolation — so only one booking succeeds.

What to remember

A — all or nothing

C — rules stay true

I — concurrent txs safe

D — commit survives crash

Architect takeaway

Use full ACID on money / stock / seats

Relax only where stale or approximate is OK

Scale often weakens A–I–D — know what you give up